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PROFESSIONAL SUMMARY

Results-driven Bioinformatics Research Associate with 18+ years of experience delivering end-to-end genomic data solutions at scale, including whole genome sequencing (WGS), RNA-seq, genome-wide association studies (GWAS), single-cell RNA-seq (scRNA-seq), and next-generation sequencing (NGS) workflows. Proven ability to process and analyze 10,000+ samples, automate quality control pipelines, and translate complex multi-omics datasets into actionable biological insights across academic and industry environments. Experienced in pipeline development using Python, R, and Snakemake, with a strong focus on reproducibility, scalability, and high-performance computing. Demonstrated success in cross-functional collaboration and training scientific teams to enhance productivity and accelerate project delivery.

CORE COMPETENCIES

- Whole Genome Analysis
- RNA-seq Analysis
- Genome Wide Association Studies (GWAS)
- Single-cell RNA-seq and scATAC-seq Analysis
- Exome Analysis
- EPIC Array Methylation Analysis
- QTL Analysis (eQTL, miR-eQTL, meQTL)
- Meta-genome & Meta-transcriptome Analysis
- Denovo-assembly of Whole Genome and Transcriptome
- NGS Data Demultiplexing & Quality Analysis (Illumina HiSeq & MiSeq)
- Population genomics (gatk, plink, admixture, nucleotide diversity)

TECHNICAL SKILLS

- Bioinformatics Programming : Python , Perl , R, , Shell , R Shiny
- Genomics: WGS, RNA-seq, GWAS, eQTL, scRNA-seq, Variant Calling
- Tools: GATK, PLINK, bcftools, samtools, Seurat, Snakemake,bwa, bowtie2, STAR, DESeq2, ANNOVAR
- Systems: HPC (SLURM), Linux
- Pipeline Development using Snakemake and Software Installation
- Data Management and Statistical Analysis
- Training and Workshop Facilitation
- Machine Learning & AI: Supervised/Unsupervised Learning, Regression, Classification, Feature Engineering, Model Evaluation, scikit-learn

EDUCATION

MBA in Business Analytics — *O.P. Jindal Global University, India (2025 – 2026)*

Relevant Coursework: Machine Learning, Artificial Intelligence, Data Science

M.Sc. in Bioinformatics — *Amrita Vishwa Vidyapeetham, India (2005 – 2007)*

Thesis: ExDom: An integrated database for comparative analysis of exon–intron structures of protein domains in eukaryotes (Published in NAR)

B.Sc. in Biotechnology — *Bharathiar University, India (2002 – 2005)*

CERTIFICATIONS & TRAININGS

- Biomedical Research: CITI Program
- Managing multiple projects (2024): LinkedIn
- Tools for data science (2024): IBM Coursera
- Introduction to Nextflow (2023): Physalia
- Introduction to Python Programming for Biologists (2023): Physalia
- Version Control for Researchers (2023): Physalia
- R Essential Training: Wrangling and Visualizing Data (2022): LinkedIn
- Fundamentals of Immunology: Innate Immunity and B-Cell Function (2021): Rice University – Coursera
- What is Data Science (2019): IBM Coursera
- MalariaX: Defeating Malaria from the Genes to the Globe (2019): HarvardX
- 6 Sigma Green Belt (2015): KAUST

PROFESSIONAL EXPERIENCE – SELECTED PROJECTS

Research Associate, ***Public Health Research Center, NYU Abu Dhabi*** *Sep 2021 – Present*

- Exploring machine learning methods to build models on the RNA-seq data for predictive insights.
- Led the quality control and analysis of 10,000 whole genome Axiom array genotyping datasets using Axiom Suite and PLINK, resulting in increased dataset accuracy for the UAE Health and Future study.
- Coordinated the sequencing of 1,000 whole genome sequences using Illumina technology, enhancing the genotyping process for the UAE Healthy Future study.
- Performed population structure and fine-scale genetic structure analysis using PLINK, identifying key genetic variations in the population.
- Developed and implemented Python programs for automated quality control processes, including data validation and error detection, reducing missing datasets and duplicate sample identification.
- Developed and deployed comprehensive GWAS pipelines for 10K UAEHFS samples.
- Installed and maintained the REDCap application, ensuring smooth data management and user support across multiple projects.
- Developed and deployed R Shiny applications, enhancing internal database management and user interaction, leading to increased data accessibility.
- Created Snakemake pipeline for rare variant analysis on all genes for lipid profiling on WGS datasets from UAEHFS using multiple tools such as Burden, SKAT and SKAT-O. Subsequently created R-Shiny app for dynamic visualization.
- Moroccan GWAS: Phenotype data processing, cleaning and database creation; WGS data analysis including PopGen.
- Mongolian GWAS: WGS data analysis including PopGen by comparing with 1000 Genome and HGDP samples.
- Trained students and staff in bioinformatics pipelines, software, and R, significantly improving their analytical skills and project contributions.

Research Technician (Bioinformatics), ***NYU Abu Dhabi, UAE*** *Jan 2016 – Aug 2021*

- Developed pipelines for RNA-seq, WGS, and epigenetic analysis.
- Conducted GWAS, variant calling, and population stratification for genomic data.
- Analyzed single-cell RNA-seq data for human (AGHJ 2025 – https://doi.org/10.1016/j.ajhg.2025.01.020).
- Performed systems genetics, miRNA-mediated regulation studies, and mRNA-QTL analysis for host response in COVID-19 (Human Genomics 2023 – https://doi.org/10.1186/s40246-023-00494-4).

Technical Specialist (Bioinformatics), ***KAUST, KSA*** *Mar 2014 – Dec 2015*

- Conducted comparative genomics and metabolic pathway analysis of Red Sea microbial communities.
- Developed automated Perl scripts for metagenomic and transcriptomic data processing.

- Transcriptome analysis of Epulopiscium fishelsoni from the gut of surgeonfish from the Red Sea (PNAS 2017 – https://doi.org/10.1073/pnas.1703070114).
- Metabolic traits of the uncultured archaeal MSBL1 lineage from the Red Sea brine pools (Nature 2016 – https://doi.org/10.1038/srep19181).

Associate Bioinformatics Analyst, ***SciGenom Labs, India*** *Feb 2011 – Jan 2014*

- Analyzed whole genome sequencing data of plant growth-promoting bacteria (PLOS ONE 2014 – https://doi.org/10.1371/journal.pone.0104259).
- Conducted downstream analysis of patient Sanger sequencing data for oncogenes and cardiology-related drug responses using tools like Sequencher and Mutation Surveyor; generated customized clinical reports, and created reference files to support future diagnostic workflows.
- Developed PrimMe, an integrated tool for generating primer sequences based on Entrez gene IDs or custom genomic coordinates, using PHP, Perl, MySQL, and Linux.

Senior Bioinformatics Programmer, ***Genome Life Science, India*** *Jul 2007 – Jan 2011*

- RoBuST (Root & Bulb Genomics Database): An integrated genomics resource for the root and bulb crop families Apiaceae and Alliaceae, developed using Perl, CGI, MySQL in Linux (https://doi.org/10.1186/1471-2229-10-161).
- ExDom (Exon Domain Database): A unique database for the comparative visualization and analysis of Exon–Intron structure of protein domains in seven eukaryotic genomes, developed using Perl, CGI, MySQL (https://doi.org/10.1093/nar/gkn746).

SELECTED CONFERENCES & POSTER PRESENTATIONS

- Bioinformatics Training for the Center for Brain Health (CBH) – 2024, UAE: Delivered hands-on workshops covering NGS data analysis, pipeline development, bioinformatics tools, and GWAS.
- Workshop: Variant Detection & Annotation – Feb. 2024, NYUAD, UAE: Hands-on workshop on generating gVCFs from raw sequencing data, annotation, and downstream interpretation.
- Bridging Gaps Between Genomics and Medicine in Africa – 2019, Morocco: Guest speaker at the 1st International Congress of the Moroccan Society of Genomics and Human Genetics.
- Winter Institute of Statistical Genetics, Abu Dhabi – 2019: Attended workshops; part of organizing committee.
- Winter Institute of Statistical Genetics, Abu Dhabi – 2017: Attended workshops; part of organizing committee.
- NextGen Genomics & Bioinformatics Technologies (NGBT) Conference – 2013, CSIR-IGIB.
- Next Generation Sequencing and Bioinformatics for Genomics & Healthcare Conference – 2012, IIT Madras, India. Presented poster on PrimeMe; part of organizing committee.
- NextGen Genomics & Bioinformatics Technologies (NGBT) Conference – 2011, RGCB, India.

SELECTED PUBLICATIONS

- Tala Shahin, Jakub Jurkovic, Mame Massar Dieng, **Vinu Manikandan**, et al. Single-cell transcriptomics reveals inter-ethnic variation in immune response to Falciparum malaria. The American Journal of Human Genetics (**2025**). doi: 10.1016/j.ajhg.2025.01.020
- Almojil D, **Manikandan V**, Drou N, Measey J, Boissinot S. The two sub-genomes of the allotetraploid frog Xenopus laevis are evolving under similar selective pressure in extant populations. BMC Genomics. (**2025)** Oct 7;26(1):887. doi: 10.1186/s12864-025-12036-4.
- Son GY, Zou A, Wahl A, ..., **Vinu M**, ..., Idaghdour Y, Yule DI, Feske S, Lacruz RS. Loss of STIM1 and STIM2 in salivary glands disrupts ANO1 function but does not induce Sjogren's disease. Function Oxford (**2025**). doi: 10.1093/function/zqae047
- El Hajj Y, Shahin T, Dieng MM, ..., **Manikandan V**, Idaghdour Y. Pregnenolone sulfate induces transcriptional and immunoregulatory effects on T cells. Scientific Reports (2024). doi: 10.1038/s41598-024-57327-0
- Son GY, Tu NH, ..., **Vinu M**, ..., Idaghdour Y, ..., Lacruz RS. The Ca2+ channel ORAI1 is a regulator of oral cancer growth and nociceptive pain. Science Signaling (2023). doi: 10.1126/scisignal.adf9535

- Almojil D, Diawara A, ..., **Manikandan V**, ..., Idaghdour Y. Impact of Plasmodium falciparum infection on DNA methylation of circulating immune cells. Frontiers in Genetics (2023). doi: 10.3389/fgene.2023.1197933
- Gjorgjieva T, Chaloemtoem A, ..., **Manikandan V**, ..., Idaghdour Y. Systems genetics identifies miRNA-mediated regulation of host response in COVID-19. Human Genomics (2023). doi: 10.1186/s40246-023-00494-4
- Wael Abdrabou, Mame Massar Dieng, ..., **Vinu Manikandan**, ..., Idaghdour Y. Metabolome modulation of the host adaptive immunity in human malaria. Nature Metabolomics (2021). doi: 10.1038/s42255-021-00404-9
- Mame Massar Dieng, Aissatou Diawara, **Vinu Manikandan**, ..., Idaghdour Y. Integrative genomic analysis reveals mechanisms of immune evasion in P. falciparum malaria. Nature Communications (2020).
- Francisco J. Aulestia, ..., **Vinu Manikandan**, ..., Idaghdour Y, Yule DI, Lacruz RS. Fluoride exposure alters Ca2+ signaling and mitochondrial function in enamel cells. Science Signaling (2020).
- Houda Chelaifa, **Manikandan Vinu**, ..., Idaghdour Y, Hasan A, Hernandez HH. Beyond the greenhouse: coupling environmental and salt stress response in Salicornia bigelovii. BioRxiv (2020).
- Guan Y, Ngugi DK, **Vinu M**, ..., Stingl U. Comparative Genomics of the Genus Methanohalophilus. Frontiers in Microbiology (2018).
- L'Haridon S, Corre E, Guan Y, **Vinu M**, ..., Jebbar M. Complete Genome Sequence of Methanohalophilus portucalensis Strain FDF-1T. Genome Announcements (2018).
- Ngugi DK, Miyake S, ..., **Vinu M**, ..., Stingl U. Genomic diversification of giant enteric symbionts reflects host dietary lifestyles. PNAS (2017).
- Eckstein M, Vaeth M, ..., **Vinu M**, ..., Idaghdour Y, Feske S, Lacruz RS. Store-operated Ca2+ entry (SOCE) controls ameloblast cell function and enamel development. JCI Insight (2017).
- Shraim R, Dieng MM, **Vinu M,** ..., Idaghdour Y, Burt JA. Environmental Extremes Are Associated with Dietary Patterns in Arabian Gulf Reef Fishes. Frontiers in Marine Science (2017).
- Jimenez-Infante F, Ngugi DK, **Vinu M**, ..., Stingl U. Genomic characterization of two novel SAR11 isolates from the Red Sea. FEMS Microbiology Ecology (2017).
- L'Haridon S, Corre E, Guan Y, **Vinu M**, ..., Jebbar M. Complete Genome Sequence of Methanohalophilus halophilus DSM 3094T. Genome Announcements (2017).
- Mwirichia R, Alam I, ..., **Vinu M**, ..., Stingl U. Metabolic traits of an uncultured archaeal lineage–MSBL1–from brine pools of the Red Sea. Nature Scientific Reports (2016).
- Jimenez-Infante F, Ngugi DK, **Vinu M**, ..., Stingl U. Comprehensive Genomic Analyses of the OM43 Clade. Applied and Environmental Microbiology (2015).
- Gupta A, Gopal M, Thomas GV, **Manikandan V,** et al. Whole Genome Sequencing and Analysis of Plant Growth Promoting Bacteria Isolated from Plantation Crops. PLOS ONE (2014). doi: 10.1371/journal.pone.0104259
- Ashwini Bhasi, ..., **Vinu Manikandan**, ..., Senapathy P. RoBuST: An Integrated Resource of Genomics Information for Plants in the Root and Bulb Crop Families Apiaceae and Alliaceae. BMC Plant Biology (2010).
- Ashwini Bhasi, Philge Philip, **Vinu Manikandan**, Periannan Senapathy. ExDom: an integrated database for comparative analysis of the exon–intron structures of protein domains in eukaryotes. Nucleic Acids Research (2009).
- Beena P.S. and **Vinu Manikandan**. Chapter 13: Applications of Next Generation Sequencing. Recent Developments in Biotechnology Vol. 5: Gene and Protein Engineering (2013).

References available on request.